

Devin Tseng

📍 Alian, Kaohsiung, Taiwan ✉ alianlbj23@gmail.com ☎ 0986 417 218 💬 alianlbj23 🔗 LinkedIn

Education

- BS National Kaohsiung University of Science and Technology**, Computer and Communication Engineering
 • project advisor: Shih-Huan Tseng
 Kaohsiung, Taiwan
 Sept 2019 – Sept 2023
- MS National Cheng Kung University**, Computer Science and Information Engineering
 Expected graduation on 2026-08-31.
 • Graduate advisor: Wen Yu Su
 • Lab: SCREAM Lab
 • Thesis: Coordinated Control of Multiple Robots Using ROS2 and a Unity-Based Digital Twin Environment
 • Exchange student at Hiroshima University
 Tainan, Taiwan
 Sept 2023 – Aug 2026

Experience

- ITRI Collaboration on OpenRB-150 Dynamixel Control System (2026)**, Developer
 Collaborated with ITRI to build a custom Dynamixel control system using micro-ROS and ROS with a self-designed dashboard.
 • Integrated embedded firmware and ROS workflows on OpenRB-150 for real-time Dynamixel motor control.
 • Implemented both embedded-side micro-ROS communication and high-level ROS topic-based control for rapid prototyping.
 • Developed a custom dashboard interface for intuitive operation and system monitoring.
 • **Tech stack:** micro-ROS, ROS2, PlatformIO, Dynamixel, OpenRB-150, Docker, Embedded
 Taiwan
 2026
- Tracked-Wheel Mobile Mechanism Design (2026)**, Developer
 Designed a tracked-wheel mobile mechanism using Fusion 360.
 • **Tech stack:** Fusion 360
 Tainan, Taiwan
 2026
- HIWIN RA605 Inverse Kinematics Environment Integration (2025)**, Developer
 Collaborated with HIWIN to integrate an inverse-kinematics environment for RA605 robotic arm control.
 • Integrated robotic arm drivers and MoveIt2 into a unified Docker environment for arm control.
 • **Tech stack:** MoveIt2, Docker, ROS2
 Tainan, Taiwan
 2025
- Coordinated Control of Multiple Robots Using ROS2 and a Unity-Based Digital Twin Environment (First-Generation, 2025)**, Thesis Author
 Built a first-generation multi-robot digital twin control environment.
 • Developed a multi-robot collaborative object handover system.
 • Built a 1:1 robot model using Fusion 360.
 • Developed a Unity-based digital twin environment synchronized with physical robots.
 • Implemented YOLO-based object detection and tracking from robot camera streams.
 • Implemented autonomous vehicle navigation.
 • Applied PyBullet for inverse kinematics computation.
 • Deployed onboard robot computation on NVIDIA Jetson.
 • Implemented automatic object grasping based on first-person camera perception.
 • **Tech stack:** ROS2, Unity, Fusion 360, YOLO, PyBullet, Jetson, NAV2, Embedded, PlatformIO, micro-ROS
 Tainan, Taiwan
 2025

RB3 Gen2 Object Recognition and Auto-Grasp Deployment (2025) , Developer Deployed object recognition and automatic grasping functions on RB3 Gen2 running Qualcomm Linux. - Implemented and optimized vision-guided grasping workflows on RB3 Gen2 hardware. Tech stack: RB3 Gen2, Qualcomm Linux, YOLO, ROS2, PyBullet, Embedded	Tainan, Taiwan 2025
Second-Generation Robot Development (2025) , Developer Developed a second-generation robot platform with UI/remote control, autonomous navigation, and vision-based perception. - Implemented robot locomotion and arm operation through both UI and remote-controller interfaces. - Deployed the robot system on NVIDIA Jetson for onboard computation. - Implemented YOLO-based object detection and tracking with autonomous navigation. Tech stack: micro-ROS, YOLO, PlatformIO, ROS2, Embedded, NAV2, Jetson	Tainan, Taiwan 2025
TAICA Digital Twin Platform and Tool Development for Robot Navigation (2024-2025) , Creator Built the robot digital twin platform for the TAICA Robot Navigation and Exploration course. - Built a 1:1 robot model using Fusion 360. - Integrated Unity and ROS for connected simulation and control. Tech stack: Fusion 360, Unity, ROS2, NAV2	Tainan, Taiwan 2024-2025
Architecture Department Tracked Robot Project (2024) , Developer Collaborated with HIWIN and National Cheng Kung University Department of Architecture on a tracked robot system. - Developed microcontroller firmware drivers for the tracked robot platform. - Implemented navigation capabilities for the tracked robot. Tech stack: Embedded, ROS2, NAV2	Tainan, Taiwan 2024
Product Development with Leadtek on Robotics (2023) , Developer Collaborated with Leadtek on robotic product development. - Developed robotic arm control functions and vehicle navigation capabilities. - Developed firmware for both robotic arm and autonomous mobile robot platforms. Tech stack: ROS2, NAV2, Embedded, Docker	Tainan, Taiwan 2023
UIC Project (2022-2023) , Research Assistant (Advisor: Shih-Huan Tseng) Developed a web-based material management system for industry-academia collaboration. - Implemented workflows for material checkout, restocking, new item registration, and historical change tracking. Tech stack: Django, Python, JavaScript, jQuery, HTML, CSS, MySQL	Kaohsiung, Taiwan 2022-2023
Merck , Summer Internship (CIM) Summer internship from 2022-07-01 to 2022-08-31, focused on system setup, analytics dashboards, and process automation in CIM. - Set up and deployed internal operational systems to support daily utility and training management workflows. - Developed analytical Power BI dashboards to monitor electricity, water, and other utility consumption. - Built an automated employee training workflow using Power Apps, Power Automate, and SharePoint, improving training assignment and tracking efficiency. - Built an automated power reporting pipeline using Python pandas to process, organize, and archive SCADA-generated CSV data. Tech stack: Python, Power Apps, Power BI, Power Automate, SharePoint	Hsinchu, Taiwan July 2022 – Aug 2022 2 months

NSTC Project (2021-2022) , Undergraduate Research Assistant (Advisor: Shih-Huan Tseng) Developed a brain activation game system for seniors through web technologies and human-robot interaction. - Applied MMSE assessment and reminiscence therapy concepts through web games and Pepper Robot interaction. - Supported synchronized physical and psychological care for older adults. Tech stack: Django, Python, JavaScript, jQuery, HTML, CSS, MySQL	Kaohsiung, Taiwan 2021-2022
MOST Project (2021-2022) , Undergraduate Research Assistant (Advisor: Min-Huang Ho) Contributed to a Tang Poetry corpus system under a MOST-funded project. - Built functions for style classification across historical periods, poem management, and file export. Tech stack: Django, Python, JavaScript, jQuery, HTML, CSS, MySQL	Kaohsiung, Taiwan 2021-2022

Teaching Experience

FUNAI Introductory Course , Teaching Assistant	National Cheng Kung University, Tainan, Taiwan 2026
FUNAI Advanced Course , Teaching Assistant	National Tsing Hua University, Hsinchu, Taiwan 2026
FUNAI Introductory Course , Teaching Assistant	National Cheng Kung University, Tainan, Taiwan 2025
FUNAI Advanced Course , Instructor	National Cheng Kung University, Tainan, Taiwan 2025
PAIA Blockly , Instructor Taught PAIA Blockly courses.	Lingzhou Elementary School, Kaohsiung, Taiwan 2025
TAICA - Robot Navigation and Exploration Workshop , Instructor	National Cheng Kung University, Tainan, Taiwan 2024
FUNAI - Advanced Course , Instructor	National Cheng Kung University, Tainan, Taiwan 2024
S.T.U.ST-ROS Robotics Course , Instructor	Southern Taiwan University of Science and Technology, Tainan, Taiwan 2024
FUNAI - Advanced Course , Instructor	National Cheng Kung University, Tainan, Taiwan 2023
Digital Twin Course , Instructor	National Cheng Kung University, Tainan, Taiwan 2023

Skills

Programming: Python, C++, JavaScript, SQL

Robotics & Middleware: ROS2, micro-ROS, NAV2, MoveIt2, rosbridge

Embedded & Hardware: PlatformIO, OpenRB-150, Dynamixel, Jetson, RB3 Gen2, firmware development

Digital Twin & Simulation: Unity, Fusion 360, PyBullet

AI & Computer Vision: YOLO, object detection, object tracking, vision-guided grasping

Data & Automation: Power BI, Power Apps, Power Automate, SharePoint, pandas, SCADA data processing